Descriptive Analysis of Educational Games on the Apple App Store

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Despite the increasing accessibility of mobile educational games, their content and distribution remain largely unexamined in scholarly research. This data mining study presents a descriptive analysis of educational games available on the Apple App Store. By systematically analyzing game characteristics this study aims to elucidate the current landscape of mobile educational games.

Method

1. Filtered for educational games on iOS (iPhone/ iPad) from **Moby Games Database** (n = 406)
2. Used App IDs from Moby to scrape metadata and the 100 newest reviews from **App Store** (n = 254)
3. Dropped Apps no longer on the App Store or App ID has changed (n = 216)

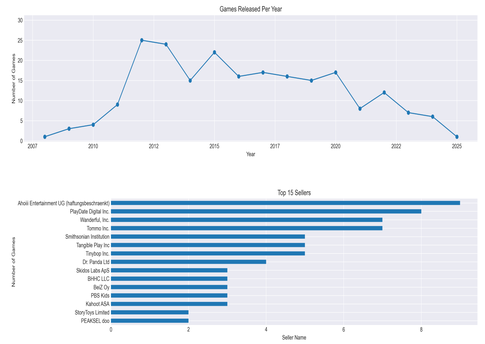
Next Steps

- Mine top 100 reviews for universal topics, themes, sentiment, etc.
- Compare common user review themes with existing evaluation rubrics of educational mobile games.

Github for poster, data, and more!

Mobile game-based learning is great!

If only there were games to play...

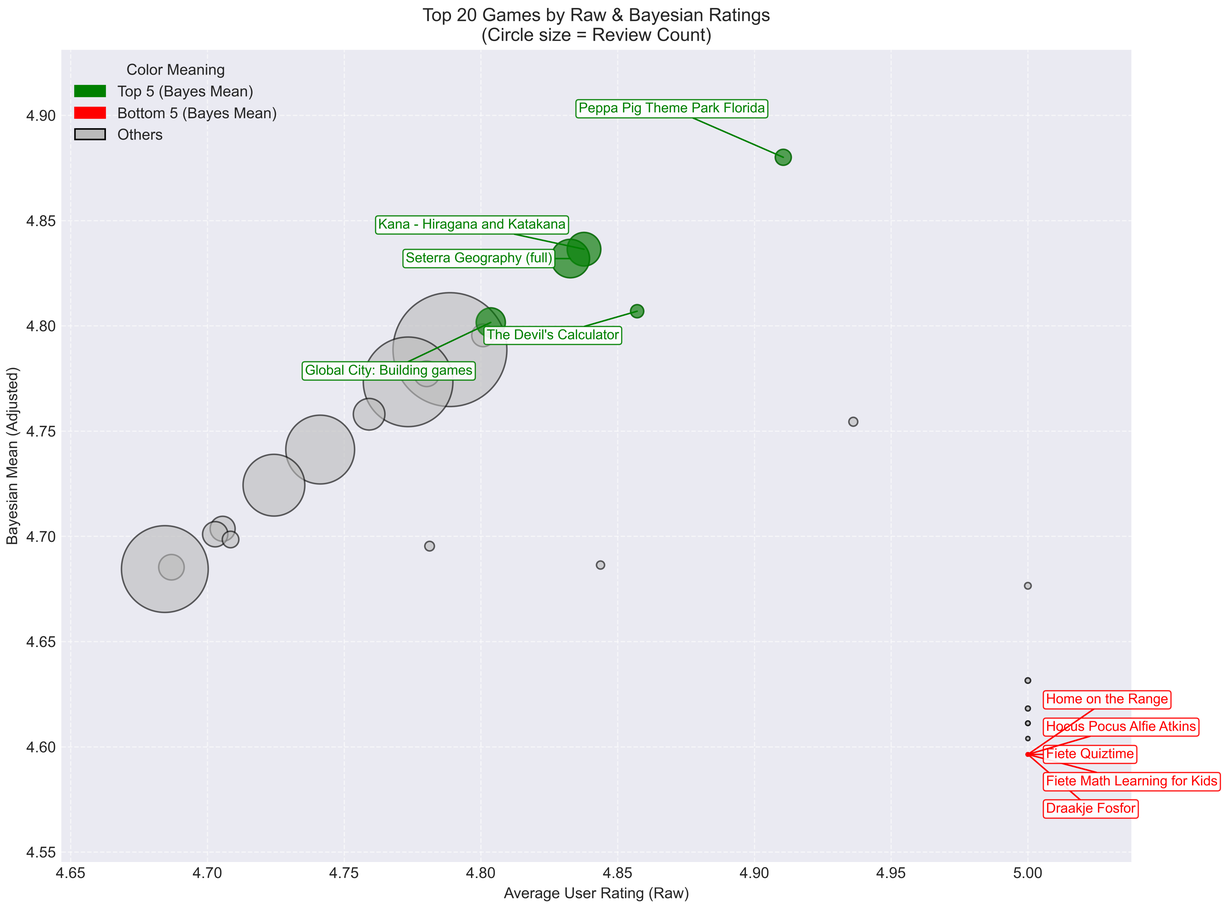


Bayesian Mean

Low-sample games are pulled toward the global mean rating (C). The median review count (m) controls how strong that pull is.

$$\text{bayes_mean} = \frac{v}{v+m}R + \frac{m}{v+m}C$$

R = app's average rating
v = number of reviews
C = global mean rating
m = "prior strength", median review count



Engagement Factor

Rewards sustained popularity, penalizes stagnant apps; capped to avoid runaway boosts

$$EI = \left(\frac{\text{reviews per year}}{\text{median reviews per year}}\right)^{\alpha}$$

